Calculus

Sub Topics

1) Limits
2) Continuity
3) Differentiability
4) Mean Value Theorems
5) Taylor & Maclaurin Series
6) Partial & Total derivatives
7) Maxima & Minima
8) Definite integrals
9) Improper integrals
10) Multiple integrals
11) Vector Differentiation
12) Vector Integration
13) Fourier Series.

Gate → 30 years papers → (2-6) M

IES → 6 year papers

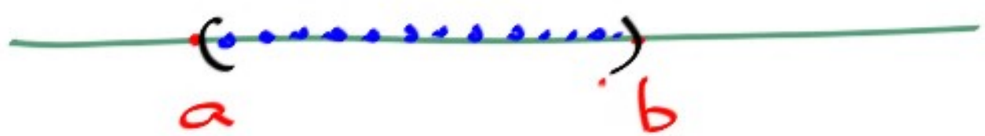
2017 → 3Q × 2M = 6M
18 →
19 →
20 →
21 → 6Q × 2M = 12M
22 → 7Q × 2M = 14Q

① Limits

(I) Intervals:

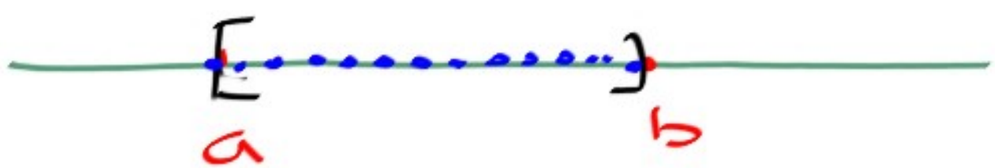
① Open interval:

$$(a,b) = \{x : a < x < b\}$$



② Closed interval:

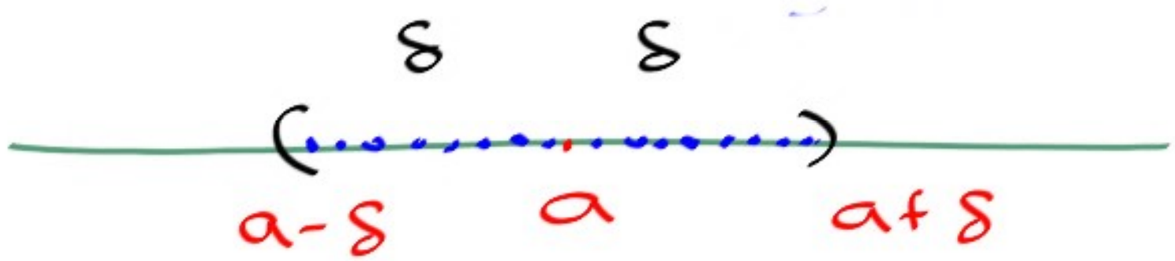
$$[a,b] = \{x : a \le x \le b\}$$



③ Neighbour-hood (nbd) of a point 'a':

If $a, \delta \in R \ni \delta > 0$ then the set of all real numbers between $a-\delta$ & $a+\delta$ is called a <u>nbd</u> of a point 'a' and it is denoted by "$N_\delta(a)$"

$$\therefore \quad N_\delta(a) = \{ x : a-\delta < x < a+\delta \} = (a-\delta, a+\delta)$$

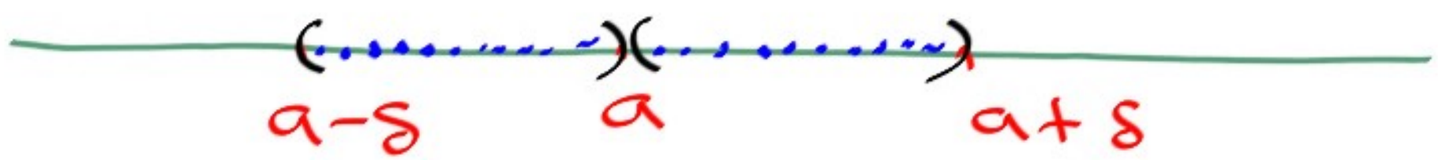


(4) Deleted nbd of a point 'a':

If $N_\delta(a)$ is a nbd of a point 'a' then the set "$N_\delta(a) - \{a\}$" is called a deleted nbd of a point 'a'.

$$\therefore N_\delta(a) - \{a\} = \{x : a-\delta < x < a+\delta \ \& \ x \neq a\}$$

$$= (a-\delta, a) \cup (a, a+\delta)$$



(II) Limits:

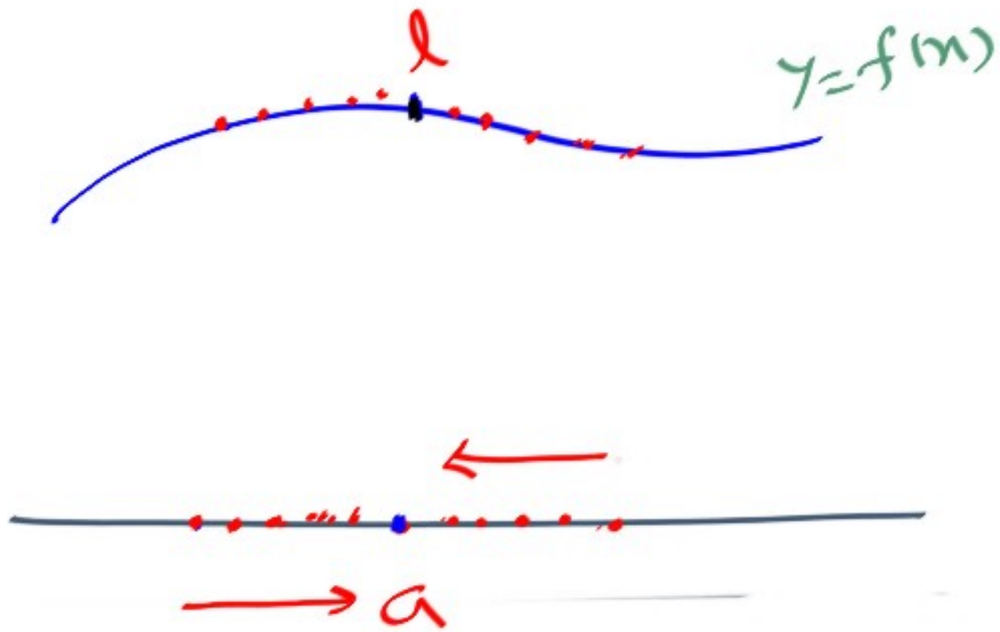
(1) Limit of a function $f(x)$ at a point $x=a$:

Let $y=f(x)$ be defined in the deleted nbd of a point $x=a$.

If all the values of the function approach a unique real number 'l' as 'x' approaches 'a' from either side then the unique real number 'l' is called the limit of the function at a point $x=a$.

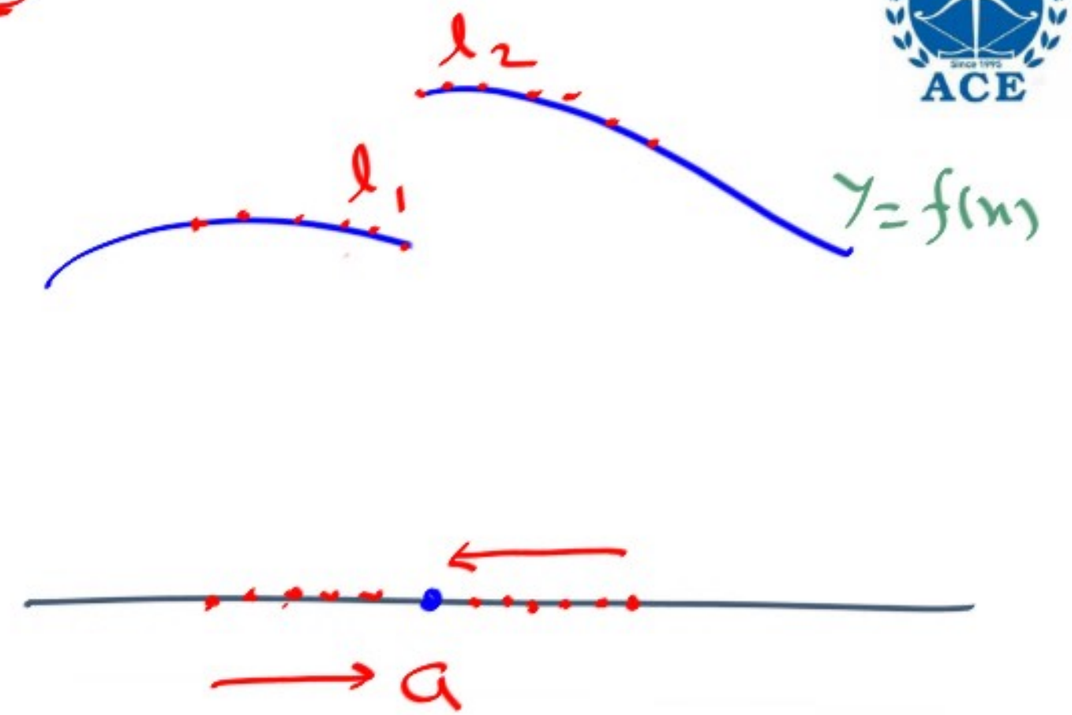
Symbolically the above can be written as $\boxed{\underset{x\to a}{Lt}\, f(x) = l}$

Note ①



Limit of a function f(x) exists at x=a.

Note ②



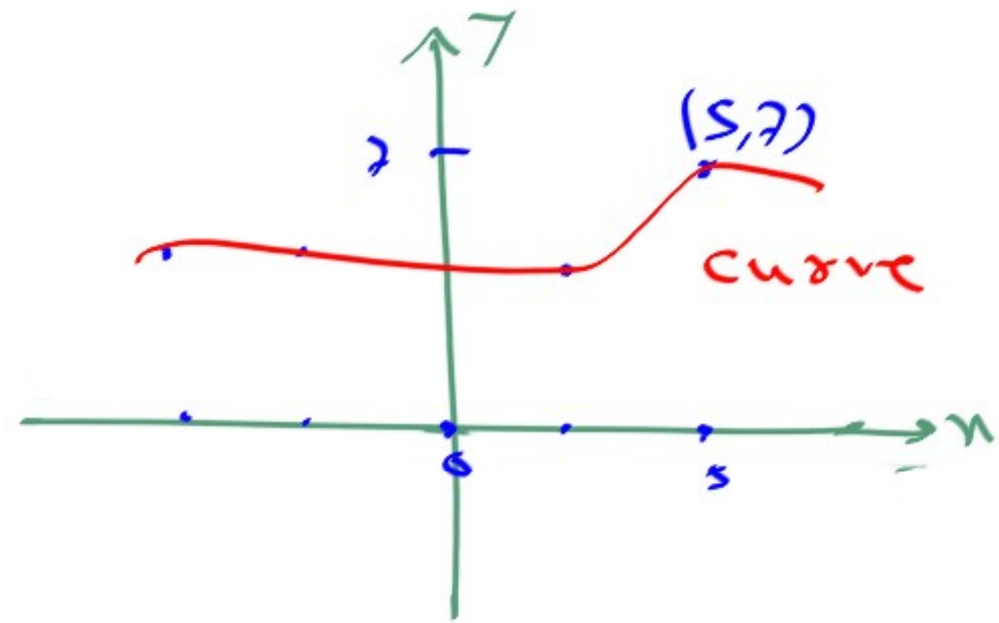
Limit of a function f(x) does not exist at x=a.

$$y = f(x) = \frac{x+2}{x-4} \longrightarrow (x, y) \text{ (or) } (x, f(x))$$

$$x = 3, \quad y = f(3) = \frac{3+2}{3-4} = -5$$

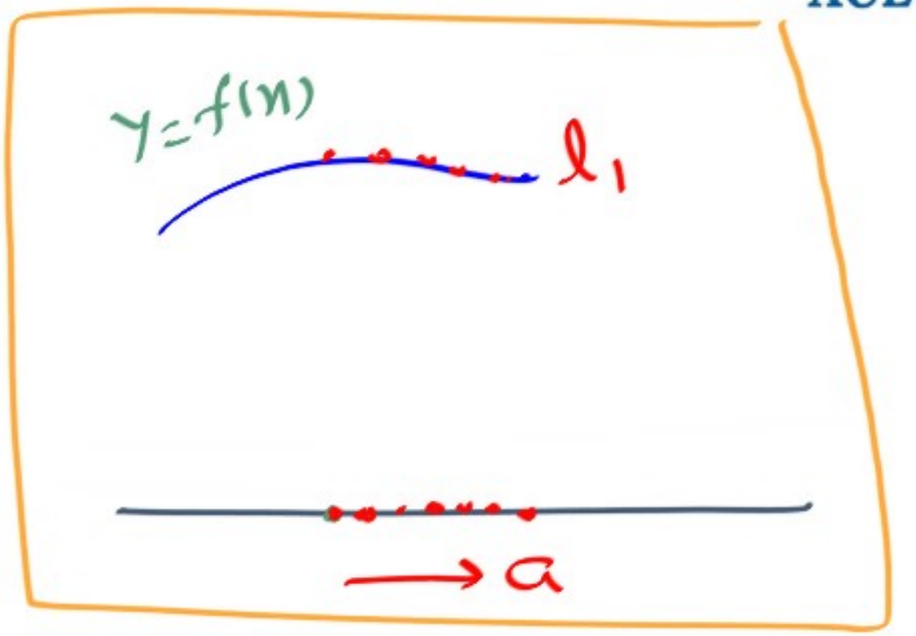
$$x = 5, \quad y = f(5) = \frac{5+2}{5-4} = 7$$

$$x = 4, \quad y = f(4) = \frac{4+2}{4-4} = \frac{6}{0} = \infty$$



② Left hand limit of a function $f(x)$ at $x=a$:

$$L.H.L = f(a^-) = \underset{x \to a^-}{Lt} f(x)$$

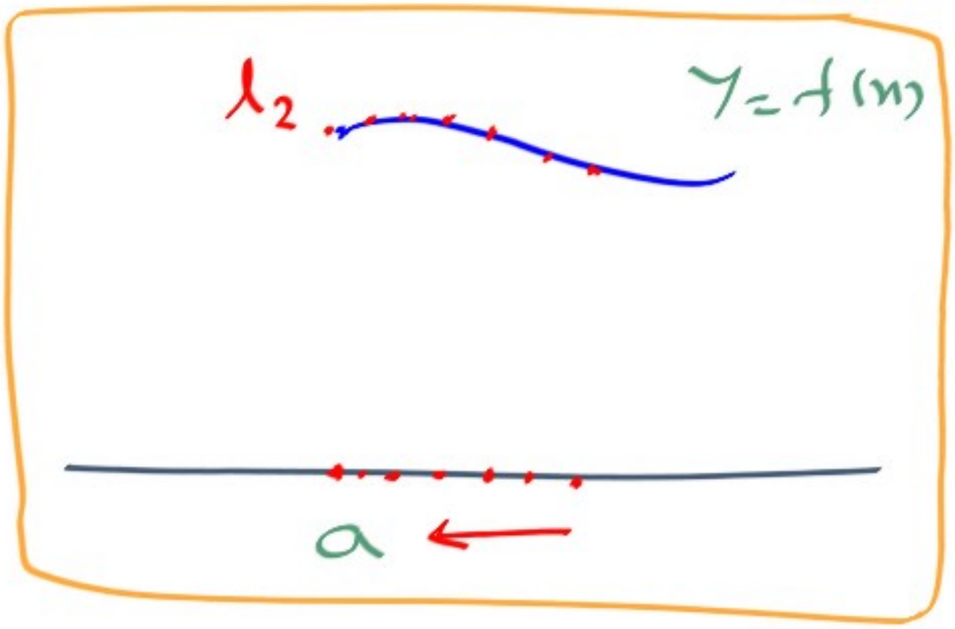


Note ①

$$\boxed{x \to a^- \implies x < a}$$

③ Right hand limit of a function $f(x)$ at $x = a$:

$$R \cdot H \cdot L = f(a^+) = \underset{x \to a^+}{Lt} f(x)$$

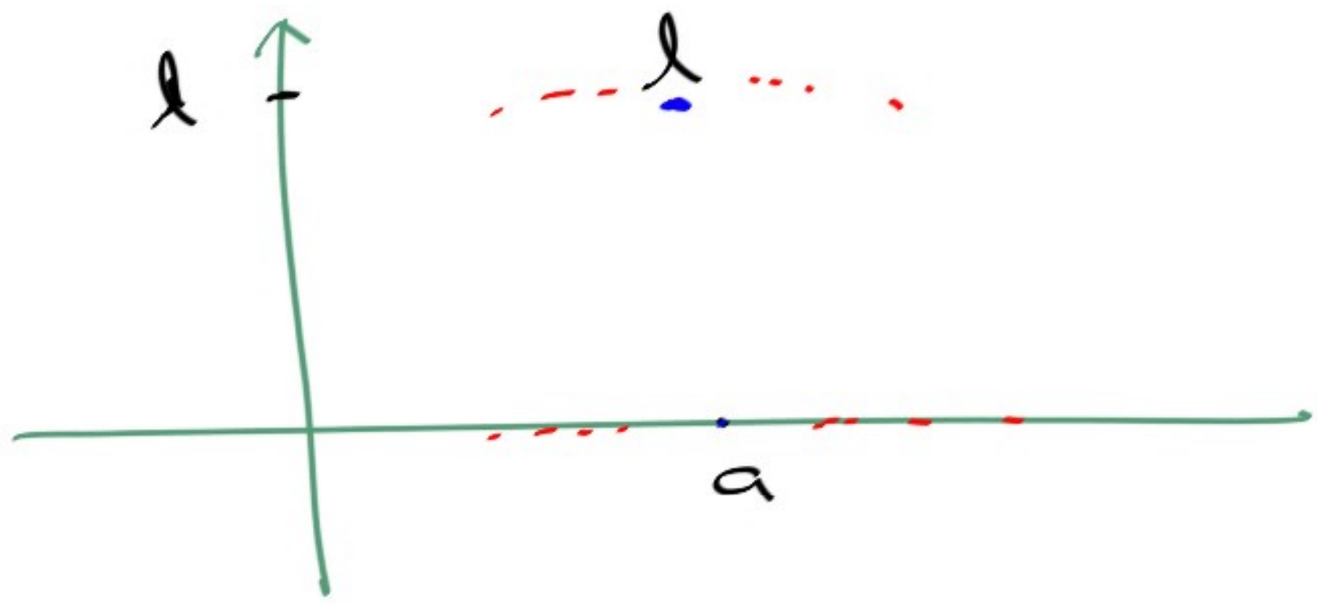


Note ①

$$\boxed{x \to a^+ \implies x > a}$$

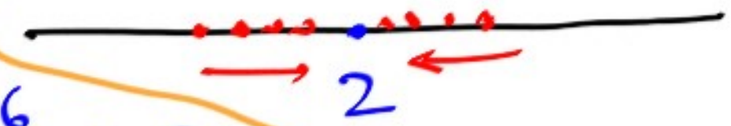
(4) Condition for the existence of limit of $f(x)$ at $x=a$:

$$\underset{x\to a^-}{Lt}\, f(x) = l = \underset{x\to a^+}{Lt}\, f(x) \iff \underset{x\to a}{Lt}\, f(x) \text{ exist } \& \ \underset{x\to a}{Lt}\, f(x) = l.$$



① $\underset{x \to 2}{Lt} \dfrac{x^2+5x+6}{4} =$

$f(x) = \dfrac{x^2+5x+6}{4}$



Sol:

$$L.H.L = f(2^-) = \underset{x \to 2^-}{Lt} f(x) = \underset{x \to 2}{Lt} \frac{x^2+5x+6}{4} = 5$$

$$R.H.L = f(2^+) = \underset{x \to 2^+}{Lt} f(x) = \underset{x \to 2}{Lt} \frac{x^2+5x+6}{4} = 5$$

Here, $L.H.L = 5 = R.H.L$ (or) $f(2^-) = 5 = f(2^+)$

$\therefore \underset{x \to 2}{Lt} f(x)$ exists & $\underset{x \to 2}{Lt} f(x) = 5$.

$$f(x) = \begin{cases} | \quad | \\ [\quad] \\ \infty \\ \{ \text{—}, (\ ,\ ) \\ \text{—}, (\ ,\ ) \\ \text{—}, (\ ,\ ) \end{cases}$$

(or) Condition for x.

$$\lim_{x \to 2} \frac{x+4}{3} = \frac{2+4}{3} = 2$$

② $\lim_{x \to 0} \frac{|x|}{x} = ___$, where $|x|$ is a modulus function.

Sol: Modulus function

$$y = f(x) = |x| = \begin{cases} x, & x > 0 \\ 0, & x = 0 \\ -x, & x < 0 \end{cases} = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

For $x < 0$, $y = f(x) = |x| = -x$

$x = -1 < 0$, $y = f(-1) = |-1| = -(-1) = 1$

$x = -2 < 0$, $y = f(-2) = |-2| = -(-2) = 2$

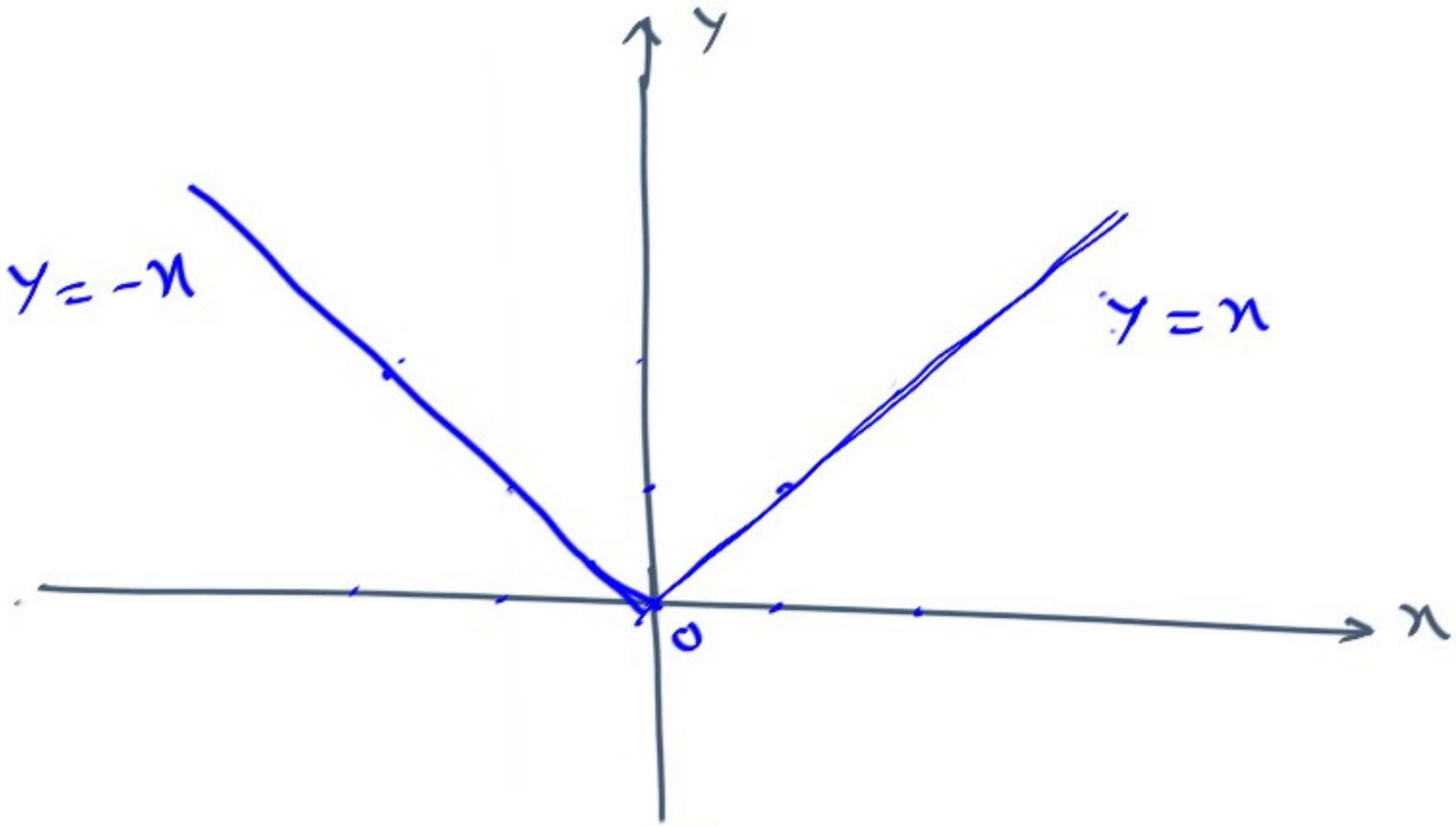
$\vdots$

For $x > 0$, $y = f(x) = |x| = x$

$x = 1 > 0$, $y = f(1) = |1| = 1$

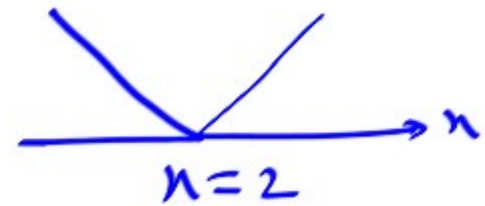
$x = 2 > 0$, $y = f(2) = |2| = 2$

$\vdots$

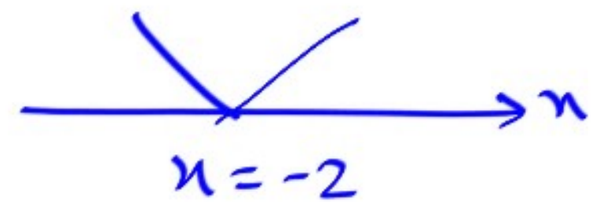


$y = f(x) = |x|.$

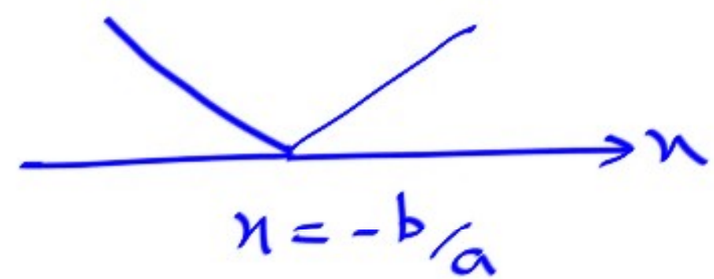
$y = f(x) = |x-2|$



$y = f(x) = |x+2|$



$y = f(x) = |ax+b|$



$$L.H.L = \underset{x \to 0^-}{Lt} \frac{|x|}{x} = \underset{x \to 0}{Lt} \frac{-x}{x} = \underset{x \to 0}{Lt} -1 = -1$$

$$R.H.L = \underset{x \to 0^+}{Lt} \frac{|x|}{x} = \underset{x \to 0}{Lt} \frac{x}{x} = \underset{x \to 0}{Lt} 1 = 1$$

Here, $L.H.L = -1 \neq R.H.L = 1$

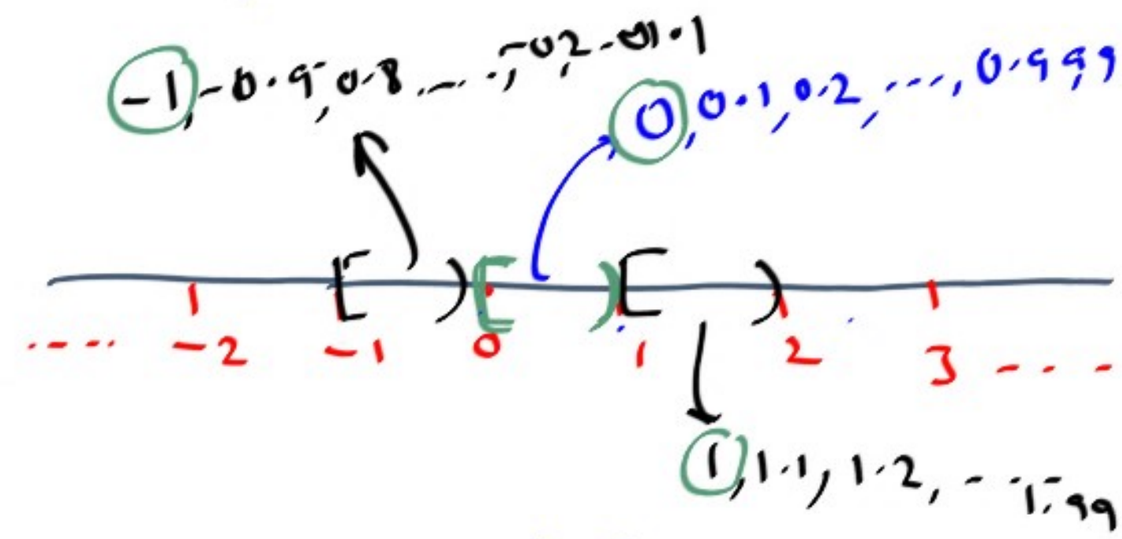
$\therefore \underset{x \to 0}{Lt} \frac{|x|}{x}$ does not exist.

③ $\underset{x \to 3}{Lt}\ [x] = -$, where $[x]$ is a step function.

Sol: Step or Bracket or Greatest integer function:

$y = f(x) = [x] = n \ \ni\ n \le x < n+1$, where $n \in I$ & $x \in R$.

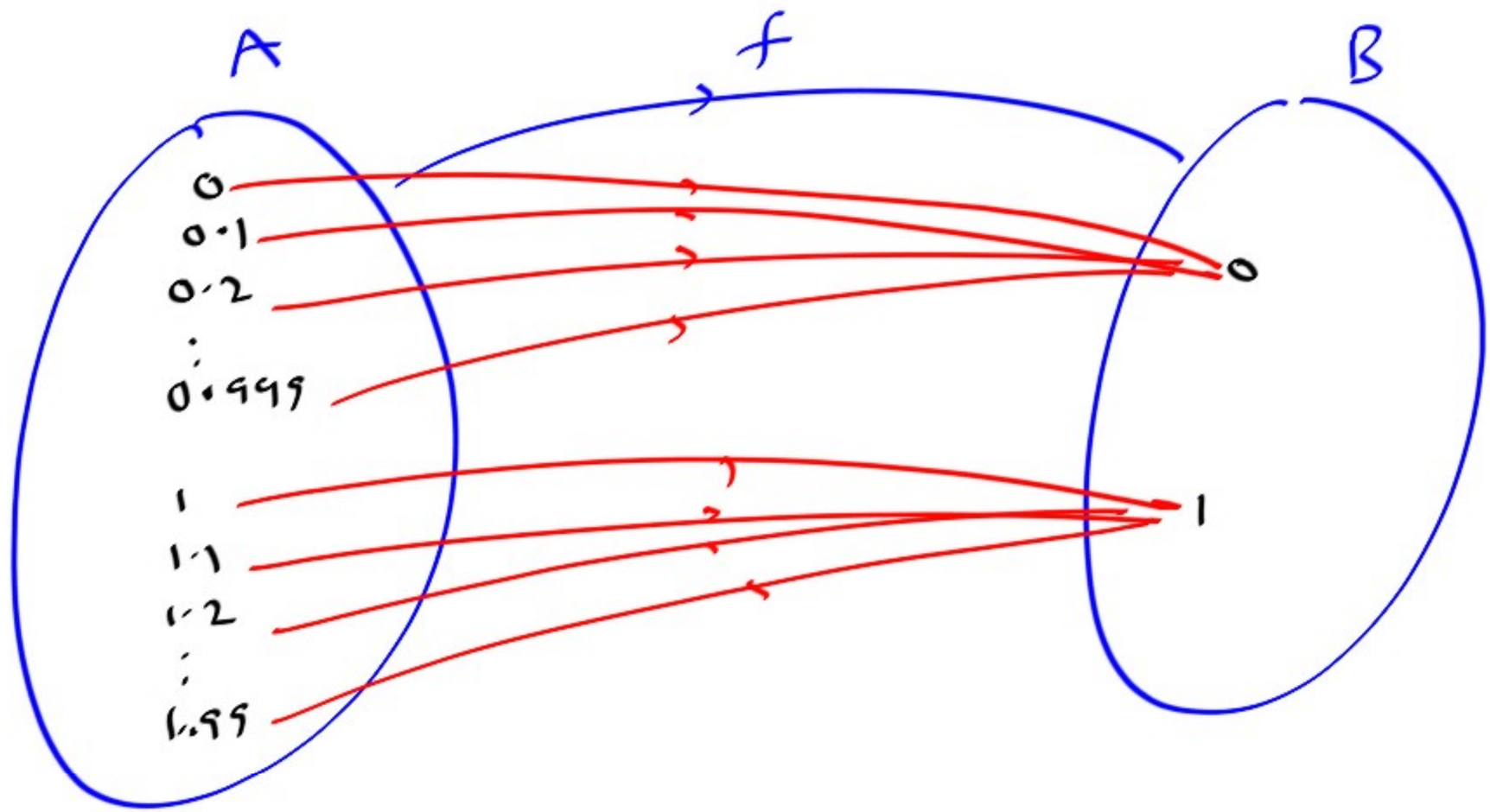
(or) $y = f(x) = [x] = \begin{cases} -2, & -2 \le x < -1 \\ -1, & -1 \le x < 0 \\ 0, & 0 \le x < 1 \\ 1, & 1 \le x < 2 \\ 2, & 2 \le x < 3 \\ \vdots & \end{cases}$

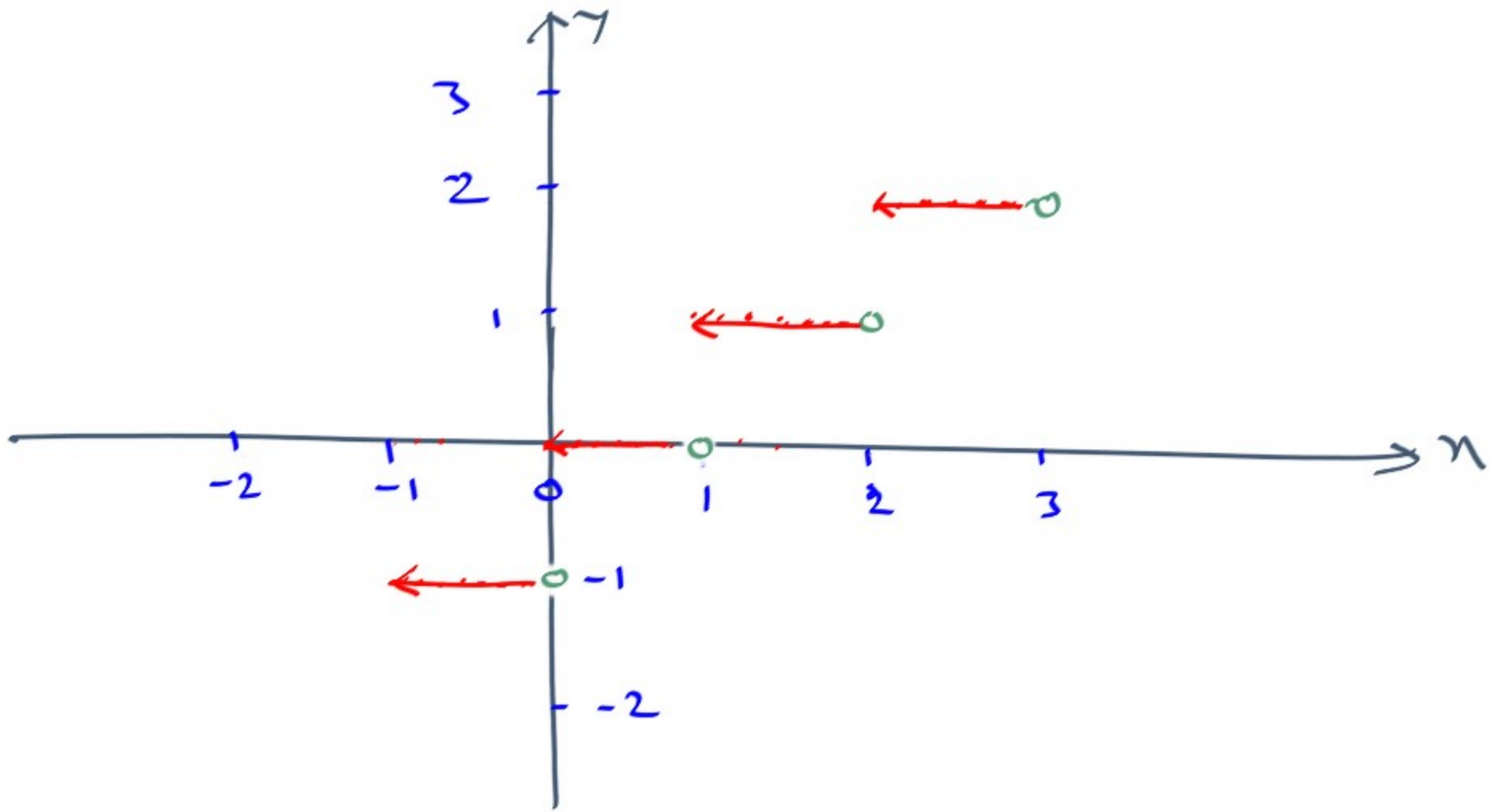


1, 2, 3 - -

0, 1, 2, 3

0, ±1, ±2, -1 -


A
f
B
0
0.1
0.2
0.999
1
1.1
1.2
1.99
0
1

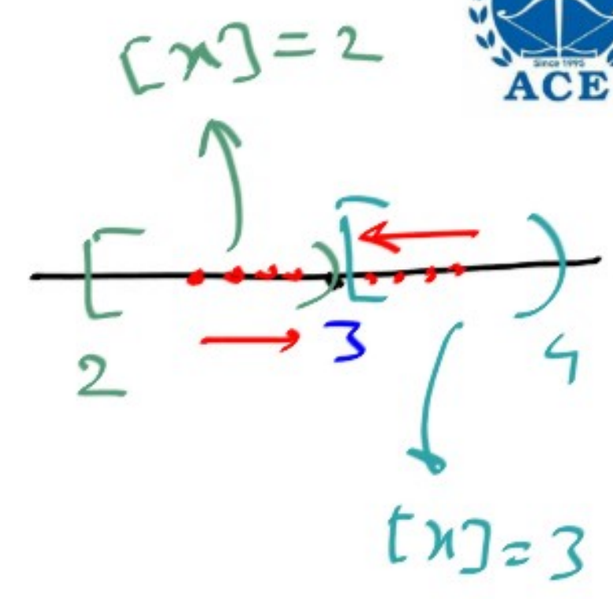

y
3
2
1
-2
-1
0
1
2
3
x
-1
-2

$$L.H.L = \underset{x \to 3^-}{Lt} [x] = \underset{x \to 3}{Lt} 2 = 2$$

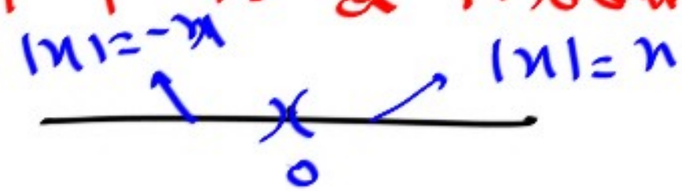
$$R.H.L = \underset{x \to 3^+}{Lt} [x] = \underset{x \to 3}{Lt} 3 = 3$$

Here, $L.H.L = 2 \neq R.H.L = 3$

$\therefore \underset{x \to 3}{Lt} [x]$ does not exist.



(4) $\underset{x \to 2}{Lt} \dfrac{|x-2|}{x-2} = -$, where $|\cdot|$ is a modulus function



Sol:- $\because$ $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

$\Rightarrow |x-2| = \begin{cases} (x-2), & (x-2) \geq 0 \quad (\text{or}) \quad x \geq 2 \\ -(x-2), & x-2 < 0 \quad (\text{or}) \quad x < 2. \end{cases}$

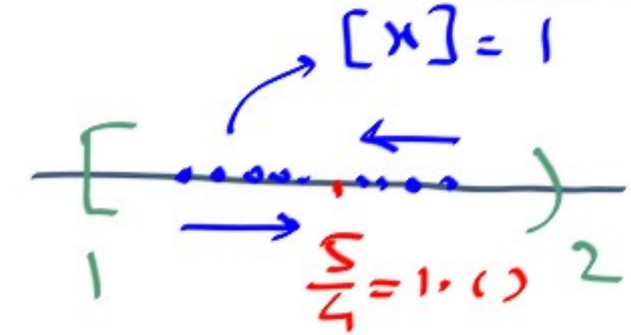
$L.H.L = \underset{x \to 2^-}{Lt} \dfrac{|x-2|}{x-2} = \underset{x \to 2}{Lt} \dfrac{-(x-2)}{x-2} = -1$

$R.H.L = \underset{x \to 2^+}{Lt} \dfrac{|x-2|}{x-2} = \underset{x \to 2}{Lt} \dfrac{x-2}{x-2} = 1$

$\therefore \underset{x \to 2}{Lt} \dfrac{|x-2|}{x-2}$ does not exist.

(5) $\underset{x \to \left(\frac{5}{4}\right)}{Lt} [x] = -$, where $[x]$ is a step function.

Sol:-



$$L.H.L = \underset{x \to \left(\frac{5}{4}\right)^-}{Lt} [x] = \underset{x \to \left(\frac{5}{4}\right)}{Lt} 1 = 1$$

$$R.H.L = \underset{x \to \left(\frac{5}{4}\right)^+}{Lt} [x] = \underset{x \to \left(\frac{5}{4}\right)}{Lt} 1 = 1$$

$$\therefore \underset{x \to \left(\frac{5}{4}\right)}{Lt} [x] = 1.$$

(6) $\underset{x \to \left(\frac{\pi}{4}\right)}{\mathrm{Lt}} \; x = __$

Sol:

$$\underset{x \to \left(\frac{\pi}{4}\right)}{\mathrm{Lt}} \; x = \frac{\pi}{4}.$$